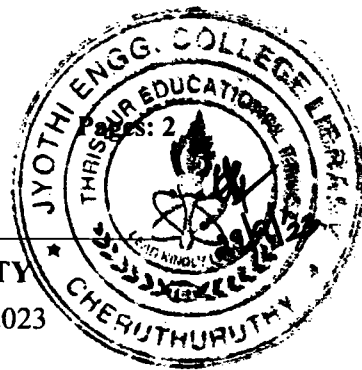




Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Second Semester M.Tech Degree Regular Examination June 2023

Discipline: COMPUTERS CIENCE AND ENGINEERING

Course Code & Name: 222TCS100 ADVANCED DATA STRUCTURES AND ALGORITHMS

Max. Marks: 60

Duration: 2.5 Hours

PART A

Answer all questions. Each question carries 5 marks

Marks

- 1 Derive the asymptotic bound for the recurrences.
 - a. $T(n) = 6T(n/3) + n$ (2.5)
 - b. $T(n) = 4T(n/2) + n^2$ (2.5)
- 2 Prove that the amortized cost of Fibonacci Heap Union operation is equal to its actual cost. (5)
- 3 Find the maximum bipartite matching of the graph $G (L \cup R, E)$, where $L = \{a, b, c, d, e\}$, $R = \{w, x, y, z\}$ and $E = \{(a, w), (b, w), (c, x), (b, y), (c, y), (c, z), (d, y), (e, y)\}$ (5)
- 4 Apply Miller Rabin algorithm to test whether 127 is prime or not. (5)
- 5 Design an approximation algorithm to find the vertex cover of a graph. Derive the approximation ratio α , of the algorithm. (5)

PART B

Answer any 5 questions. Each question carries 7 marks

- 6
 - a. Derive the amortized cost of stack operation including multi-pop in a sequence of m operations which contains n push operations. (4)
 - b. Apply RABIN-KARP algorithm to search for the pattern $P = "25"$ in text $T = "23893547"$, where $q = 64$. (3)
- 7
 - (a). Construct a Binomial heap using the numbers 10, 4, 9, 15, 22, 30. (3)
 - (b). Demonstrate the difference in the procedure of DECREASE-KEY operation in Fibonacci heap and Binomial heap. (4)

- 8 a. Define Binomial Tree. What is the number of binomial trees in a binomial heap with n nodes? Justify your answer. (4)
- b. Show that the amortized time of an operation is $O(n)$ if no heuristics is applied while performing union operation on a disjoint set implemented using linked list. (3)
- 9 a. Consider a flow network with source s and sink t . Prove that $f(s,V) = f(V,t)$. (2)
- b. Demonstrate FORD- FULKERSON method on a graph to find the maximum flow. (5)
- 10 (a) Differentiate Monte Carlo and Las Vegas algorithms with suitable examples. (4)
- (b) Derive the expected running time of randomized Quick sort algorithm. (3)
- 11 Verify the correctness of the matrix multiplication, $AB=C$ where, (7)
- $$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 6 & 7 \\ 9 & 10 & 11 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 6 & 7 \\ 8 & 9 & 10 \\ 11 & 12 & 13 \end{bmatrix} \quad C = \begin{bmatrix} 54 & 60 & 66 \\ 158 & 168 & 186 \\ 246 & 132 & 306 \end{bmatrix}$$
- 12 (a). Demonstrate the approximation algorithm to find the minimum cost tour (TSP) on graph. (5)
- (b). Compare the cost of the algorithm to the optimal cost. (2)
